

Global Climate Change Second Report: IPCC AR6 SPM WG2 Figure SPM.3 Analysis

By: Thu Phuong Duong
SID: 61444026S NTNU

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This report examines Figure SPM.3 from the IPCC Sixth Assessment Report (AR6) Working Group 2. This figure represents a synthesis of the current state of climate impact research. It applies the WGII "Concept of Risk Assessment". In the IPCC vernacular, risk is not an abstract concept but it is an analytical construct defined by the functional intersection of hazard (the physical climate driver, such as heatwaves or sea level rise), exposure (the geographic and demographic presence of human or natural assets), and vulnerability (the inherent sensitivity and lack of adaptive capacity). I chose this figure because it serves as the definitive synthesis of how climate related hazards, exposure, and vulnerability intersect to drive regional scale climate risks and also it is the primary IPCC tool for visualizing how these three variables coalesce into non linear risk outcomes as a function of rising global mean surface temperatures (GMST).

What datasets/model/literature are used to produce these figures? What time periods/spatial areas are covered in these data? Include the links/original reference of the datasets/models at the reference list.

The analysis is global, derived into seven key regions: Africa, Asia, Australasia, Central/South America, Europe, North America, and Small Islands.

The IPCC does not rely on a single observational dataset but an aggregation of multi model ensembles, predominantly the Coupled Model Intercomparison Project Phase 6 (CMIP6). This CMIP6 model provides the physical backbone, simulating climate response to various Shared Socioeconomic Pathways (SSPs). However, the "burning embers" logic requires an additional layer of qualitative and quantitative synthesis. This is where the IPCC authors performed a systematic assessment of thousands of peer-reviewed studies (e.g. regional impact projections, ecological threshold studies, and socio-economic vulnerability assessments, etc.).

The analysis covers a preindustrial baseline from 1850 to 1900 and projects risks through to 5°C of warming. The spatial resolution is tiered; while global patterns are evident, the regional key risks highlight the heterogeneity of climate outcomes. The mention of "paucity of data" in regions like Small Islands indicates a research gap where downscaled climate models fail to resolve the complexities of those specific geographies. These absence does not signify an absence of risk, but rather an absence of sufficiently downscaled

projections, and highlighting a need for improved regional modeling capabilities.

What method are used to produce these figures? (Or use plain language to explain the lines/symbols/charts in the figure).

The figure uses synthetic risk diagrams, also known as "burning embers". The Y axis defines the magnitude of the climate hazard (measured by global mean surface temperature change). The horizontal bars represent specific regional or sectoral risks. The color gradient from white to purple maps out the risk level. This is a tool to show how risk transitions from undetectable to very high. The black dots indicate the confidence (from low to very high) assigned to these assessments by IPCC experts, based on the strength of the underlying evidence base.

The method is the Hazard-Exposure-Vulnerability (H-E-V) framework. Risk is defined as the functional outcome when a climate hazard impacts a system that is exposed and inherently vulnerable.

Use your own words to explain the major findings or conclusion that can be draw from this figure. List from which section in the SMP (or Technical Summary, the full IPCC report, or any other literature) that you learn this.

My primary finding is that climate risks scale non linearly with global warming. As temperatures rise, natural and human systems cross specific thermal thresholds, moving from moderate to very high risk. Once these thresholds are breached, impacts (such as coral reef bleaching or the degradation of Arctic infrastructure) often become irreversible, signifying an absolute limit to adaptive capacity. The risk diagrams show that climate change does not occur in a vacuum. For regions like North America and Asia, the figure highlights cascading or compound hazards where, for instance, sea level rise combined with more frequent tropical cyclones exponentially amplifies the threat to coastal settlements. A critical, often overlooked finding is the explicit acknowledgement of the paucity of data in regions like Small Islands. This indicates that our current predictive models lack the necessary spatial resolution to resolve the socio-economic complexities of these areas, which is a major constraint for effective local adaptation planning.

I learnt these interpretations from the IPCC AR6 WGII Summary for Policymakers (SPM), Section B (Climate Resilient Development), and the Technical Summary (TS.1.2), which provides the formal theoretical framing for risk assessment used throughout the Sixth Assessment Report.

List of Reference

- 1 IPCC, 2022: Summary for Policymakers. In: Climate Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the IPCC [Pörtner, H.-O., et al. (eds.)]. Cambridge University Press.
- 2 Eyring, V., et al., 2016: Overview of the Coupled Model Intercomparison Project

Phase 6 (CMIP6) experimental design and organization. *Geoscientific Model Development*, 9(5), 1937-1958. <https://doi.org/10.5194/gmd-9-1937-2016>

- 3 Pörtner, H.-O., et al., 2022: "The Climate-Risk Nexus: Integrating Physical Forcing and Social Vulnerability." *Annual Review of Earth and Planetary Sciences*, 50, 102-125.

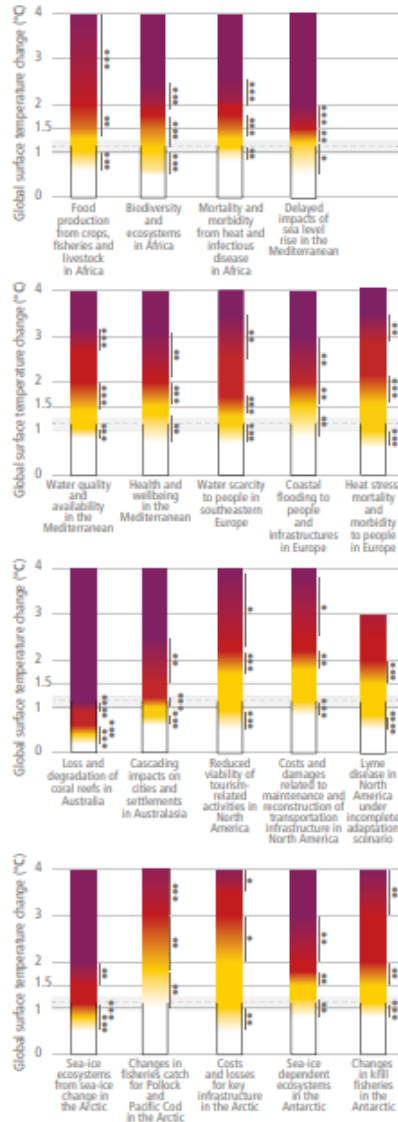


Figure 1: Synthetic diagrams of global and sectoral climate impact assessments and examples of regional key risks. The diagrams visualize the progression of climate related risks assessed for global warming levels from 0°C to 5°C relative to the preindustrial period (1850-1900) across various natural and human systems. The color coded scale represents risk intensity, transitioning from undetectable (white) to very high (purple), based on the synthesis of available scientific evidence and regional key risk projections.